

# Workplan: demand-sensitive prices from the wage block (segmented wage setting)

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**Version 2 (September 2026)** **Version 1** (September 2026)

This is the implementation annexe to ADR-0021 and to the segmented-wage ADR it calls for. It lives in `docs/` at the operator's request, as the executable companion to those decisions: the ADRs carry the choice, this file carries the steps, the tests and the decision points for the session that executes it. Nothing here is a decision, and nothing here has been run.

## 1 The goal, and what it excludes

Goal: **demand-only scenarios in which real prices respond to demand**. The paper's shocks stay where they are (the programme impulse, the F1 tilt, the F2 tax, the F3 external financing); no supply-side shock is introduced. The price response is to be generated inside the wage block instead.

Excluded, with reasons: supply-side shocks (awkward in a demand-shock framing, recorded thematically as `docs/ideas/IDEA-0001-climate-productivity-shocks.md`), imperfect competition and markups (a different price block), a second production factor (a different model), and the retired allocation wedge (never derived as a coefficient of the model).

## 2 The measurement that defines the problem

From the `matrix_5x3_v6` manifests, the price response of every labour row:

| Row                          | $\max \text{abs}(p - 1)$ (F1 / F2 / F3) | GDP deflator                   | Wage                     |
|------------------------------|-----------------------------------------|--------------------------------|--------------------------|
| BF (eta = 0, sectoral wages) | 0.221 / 0.279 / 0.279                   | 1.006326 / 1.007181 / 1.007181 | solved, 0.9734 to 1.5249 |
| ALPHA, BETA (eta = 1)        | 2.7e-15 / 2.7e-15 / 2.7e-15             | 1.000000                       | 1.000000                 |
| GAMMA, DELTA (fixed wage)    | 1.9e-13 / 2.2e-15 / 2.2e-15             | 1.000000                       | 1.000000                 |

Two conclusions. First, the `eta = 0` row is already a demand-sensitive-price row: its sectoral wages are solved given the demand composition, so prices and the deflator move with the financing cell. The paper can say something about prices in the demand-only design today, from that row, with no new machinery. Second, in every other row prices are pinned to the baseline to machine precision, and the reason is structural rather than numerical: with one factor, constant returns and a demand-free price block,  $p_i = \text{cost}_i(p, w, ip)$  depends on the wage and on intermediate prices only. Demand can reach prices **only through an endogenous wage**. That is why the fix belongs in the wage block, and it also explains why BETA and DELTA coincide with their neighbours: with prices fixed, the labour-supply elasticity has nothing to bite on and the CES-versus- Leontief contrast disappears.

One clarification of the earlier option list, because it matters for scoping: a pinned wage **vector** (Door 1, ADR-0021) produces price *heterogeneity* — a tilted pin gives  $\max \text{abs}(p - 1)$  of order 0.1 — but not price *sensitivity*: the pin is exogenous, so the price block still never sees demand. Door 1 is the complement (it makes the allocation margin observable); the route to demand-sensitive prices is a **free** wage, which is Door 3.

### 3 The specification to implement

Partition the sectors into a sticky set  $S$  (wages pinned) and a flexible set  $F$  (one common wage  $w_f$ , solved). Unknowns  $[p(1 : N); y(1 : N); w_f; F_{ext}]$ , that is  $2N + 2$ ; equations:

- $N$  zero-profit conditions, with the sector wage  $w_i = \bar{w}_i$  for  $i \in S$  and  $w_i = w_f$  for  $i \in F$ ;
- $N$  goods-market clearings, all of them, as in every regime since ADR-0019;
- one labour-market condition for the flexible segment,  $\sum_{i \in F} L_i^{cm} = \bar{L}_f \cdot ((w_f/P)/(\bar{w}_f/\bar{P}))^{\eta_s}$ , so the sticky segment's employment stays demand-determined and uncapped, as in the fixed-wage regime;
- one numeraire,  $P^{CPI} = 1$ .

Properties to rely on, and to check rather than assume:

- **Nesting.**  $S = \emptyset$  reproduces ALPHA (and BETA at  $\eta_s = 0.5$ );  $S = \text{all sectors}$  reproduces GAMMA, where the free wage disappears and the external unknown is dropped. Three existing closures become endpoints of one family, which is the presentational prize: GAMMA stops being a row and becomes the “all pinned” corner.
- **Exogenous parameter and endogenous outcome.** Only the sticky set is exogenous; the *level* of the pinned wages is a numeraire (a uniform rescale of  $(\bar{w}_S, w_f)$  leaves the real allocation invariant, as measured for the pure pin in `probe11`). The **segment wage gap**  $w_f/\bar{w}_S$  is therefore an equilibrium outcome, not a knob — which is the result worth reporting.
- **Demand-sensitive prices.**  $w_f$  appears in the clearing block through  $L^{cm}$ , so it responds to the demand composition, and with it every price through the intermediate block.
- **Determinacy.** The ADR-0017 round-gain criterion applies to the fully pinned system and does **not** carry over unchanged once a wage is free; the free wage plausibly supplies the feedback the fully pinned system lacks, so the admission question should relax — but this must be measured, not asserted.

### 4 Reconciliation with the five-door menu of docs/ETAs.md [added v2]

`docs/ETAs.md` derives the same obstruction this workplan starts from — Lemma 1 of `paper/equivalence.tex` (the price block is homogeneous of degree one in  $(p, w)$  and demand-free, so  $p = w = \text{CPI} = 1$  under  $A = 1$ ) and Corollary 1 (the real wage is pinned at its anchor) — and ranks the doors that lift it. Four consequences for this workplan, none of which changes its priority.

- **The labour door is candidate 1 there too.** The sequencing note in `docs/ETAs.md` keeps sector-specific wages as candidate 1 for the published matrix, ahead of the utilization/Verdoorn arm and the terms-of-trade arm, with the supply-side arm first only because it is cheap and earns the `eta_s` identification claim. This workplan pursues the labour door, which is the only door that repairs the demand-only matrix; the supply arm is parked for framing reasons (`docs/ideas/IDEA-0001-climate-productivity-shocks.md`). The deviation from ETAs' ordering is deliberate and is recorded here rather than taken silently.
- **Specification: general or restricted.** `docs/ETAs.md` specifies the labour door as  $N$  sectoral labour markets with a vector of sectoral supply elasticities; the specification above is the restricted variant (a sticky set plus one free wage), chosen because it nests ALPHA and GAMMA. The two are different systems: the general one makes every sectoral wage demand-sensitive and its all-rigid corner ( $\eta_{s,i} = 0$  everywhere, so  $L_i^{cm} = \bar{L}_i$ ) reproduces the `eta = 0` row, but it does not nest ALPHA or GAMMA; the restricted one nests those two rows but makes only one wage demand-sensitive. **Recommended: implement the general closure**, with the scenarios reported at a low-dimensional parametrisation (a sticky group at  $\eta_{s,i} = 0$  and a flexible group at a common elasticity), because  $N$  elasticities are weakly identified under demand-only shocks. The restricted variant then becomes a presentation choice rather than a separate implementation.
- **The Verdoorn arm is the companion, not a rival.** The capacity door's reduced form — an endogenous utilization externality in the unit-cost hook, with  $\delta < 0$  giving the Kaldor-Verdoorn case — is the cheapest door and the only one with a *negative* price sign, so running it alongside the labour door brackets the price response instead of predicting one direction. It is recorded, with the markup, capacity and external doors, as `docs/ideas/IDEA-0002-five-doors-demand-sensitive-prices.md`.
- **One schema change serves both routes.** The per-cell parameter source that `docs/ETAs.md` records as the blocking item of its own workplan (the harness hard-codes a null shock and exposes no per-cell parameter) is the same change this workplan needs for the utilization arm and for the parked supply route. It should be done once — and the identifier should be checked first: the id ADR-0018 cited in `docs/ETAs.md` is already taken by the real-GDP measurement decision, so a supply-shock schema ADR must take the next free number.

## 5 Steps

- **Step 1, prototype (no src/ change).** Write `experiments/probes/probe12_segmented_wages.jl`, mirroring the fixed-wage residual for the sticky sectors and the mobile residual for the flexible ones, reusing `_mobile_market_demand` with a full per-sector wage vector (it accepts one since ADR-0020). Measure, on the full-71 calibration: the endpoint nesting against the committed ALPHA and GAMMA manifests; the demand sensitivity of prices across the three financing cells; the segment wage gap and employment as the sticky share rises; the identity gap through the price-weighted clearing residual (the proxy validated in `probe11`, ratio 1.0000 against the kernel canary along quantity perturbations); the Jacobian conditioning and the residual floor. Record the falsifiable predictions before running: prices move with the financing cell; the wage gap widens with the sticky share; employment absorbs more as the sticky share rises.
- **Step 2, ADR.** Draft the segmented-wage ADR (proposed) with the specification above, the nesting table, the measured demand sensitivity and the admission check. Registry: a new labour-closure entry, with `labor.GAMMA` pointing at it as the all-pinned endpoint if the operator decides to present the family instead of the row.
- **Step 3, promotion.** Dispatch-only kernel change: a segmented closure type carrying the sticky set and the pinned vector, its residual function, the `solve` plumbing, and the vector-wage handling in `equilibrium_residuals`, `market_clearing_residuals`, `external_balance_canary` and `gdp_components`. The harness needs a third gate for the segmented regime (the flexible segment's labour-market residual) and, as at  $\eta = 0$ , the scalar wage metric should become the wage-bill-weighted aggregate with `wage_min` and `wage_max` carrying the structure.
- **Step 4, tests.** The endpoint nesting on the fixtures and on one full-71 cell; the level invariance under a uniform rescale; the demand-sensitivity assertion (the price deviation must move with the financing cell, which no existing closure satisfies); the identity gate; the determinacy check.
- **Step 5, design and generation.** A design carrying the existing fifteen cells plus the segmented cells (sticky share at several levels, each across F1, F2, F3), preregistered before execution. A `src/` change supersedes the current generation's provenance, so the existing cells are re-run in the same batch rather than mixed with it.
- **Step 6, records.** Flow table from the new manifests, the assessment block on demand-sensitive prices, the registry and log entries, the board, the gate.

## 6 Decision points for the operator

- **The sticky set.** Which sectors are pinned, and by what rule. This is the research choice, and it should be preregistered; a rule ("the programme sectors", "the largest half by baseline employment") is preferable to a hand-picked list, because a hand-picked list invites the tuning objection.
- **The flexible segment's supply elasticity.**  $\eta_s = 0$  makes the family nest ALPHA at one end;  $\eta_s = 0.5$  nests BETA. Either is defensible; the choice determines what the paper can say about the supply side.
- **GAMMA's role.** Keep it as a matrix row for continuity, or present the family with GAMMA as its all-pinned endpoint. The second is cleaner and answers the redundancy question, but it changes the paper's table structure.
- **Scope.** Whether the rigidity share becomes a headline dimension of the revision or a follow-up paper. Nothing in steps 1 to 3 commits either way; the probe is worth doing regardless.

## 7 Risks and how they surface

- **The mirrored prototype.** The  $\eta = 0$  work showed that a correct prototype can still be mistranslated into the kernel, where the defect is structural and silent. The endpoint nesting is the self-diagnosing check: if  $S = \emptyset$  does not reproduce ALPHA and  $S = \text{all}$  does not reproduce GAMMA, the translation is wrong.
- **Determinacy.** Measured in step 1, not assumed; if the free wage does not stabilise the system, the family is admitted only on a restricted sticky set.
- **Provenance.** A new generation is required; the existing cells must be re-run in the same batch and reproduce their manifests.
- **Interpretation.** The sticky set is exogenous and the wage gap is an outcome. A welfare difference across sticky shares is not a statement about wage policy, and the rigidity share does not identify a labour-supply elasticity.

## 8 Evidence base

`experiments/probes/probe11_gamma_wage_structure.jl` (level invariance, the identity proxy and its validation, conditioning); `probe7_sectoral_wages_eta0.jl` (the  $\eta = 0$  closure the vector-wage apparatus comes from);

probe8\_promotion\_verification.jl and probe9\_nonbf\_reproduction.jl (the promotion's verification pattern); runs/matrix\_5x3-v6-\* (the price table above); docs/VariationinGamma.md (the four doors and the claim-by-claim provenance); ADR-0020 and ADR-0021; the eta\_s identifiability item in docs/DOCS\_ASSESSMENT.md.

## 9 Revision Log

- **Version 1** (September 2026)
- **Version 2** (September 2026) — Added the reconciliation with the five-door menu of docs/ETAs.md: the labour door is that document's candidate 1 as well, so the priority stands; the specification choice between the general sectoral-labour-market closure and the restricted sticky-set variant is flagged with a recommendation (general, reported at a low-dimensional parametrisation, because N sectoral elasticities are weakly identified under demand-only shocks); the Verdoorn arm is recorded as the cheap companion with the opposite price sign; and the shared per-cell parameter source is identified as one schema change serving both routes, with the note that the ADR-0018 id cited in docs/ETAs.md is already taken.